

Supplemental File for the Key "Compared" Similarity Measures

Considering we have more than 220 similarity metrics in the framework, only the most important ones, whose performance has been highly competitive based on our trials, are succinctly presented, including the hybrid and combined measures. The hybrid and combined measures are introduced in the following section. Finally, the proposed similarity measures are drawn in the last part of this section.

The first measure, which is described here, is Jaccard because of its special feature when it does not concern the magnitude of numeric rating values. The Jaccard measure is the ratio of the cardinality of common set $I_1 \cap I_2$ to the cardinality of the union set $I_1 \cup I_2$. It measures how many common items both users rated, which is defined as given in equation 10 [8]:

$$\text{Jaccard}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \quad (10)$$

Based on the idea of singularity, [30] suggested an enhanced version of Jaccard. For instance, if rating values vary from 1 to 5, the median is $r_m = (1+5) / 2 = 3$; hence, rating values; that are greater than or equal r_m are called relevant ratings and the others that are smaller than r_m are called non-relevant. Liang (2015) et al. [30] called "relevant" and "non-relevant" as "positive" and "negative," respectively. Given user 1 and user 2, for every item j , let:

- If both r_{1j} and r_{2j} are positive, item j belongs to the predefined set PA .
- If both r_{1j} and r_{2j} are negative, item j belongs to the predefined set NA .
- If r_{1j} and r_{2j} are different, which means that one is positive and the other is negative, item j belongs to the predefined set D .
- When one user rated item j but the other did not rate it, if such a rating is positive, then item j belongs to the predefined set PO ; otherwise, item j belongs to the predefined set NO .

Meanwhile, Liang et al. (2015) defined singularity quantities such as positive singularity, negative singularity, and empty singularity denoted S_j^P , S_j^N , and S_j^E as follows:

$$\begin{aligned} S_j^P &= 1 - \frac{|P_j|}{|U|} \\ S_j^N &= 1 - \frac{|N_j|}{|U|} \\ S_j^E &= 1 - \frac{|E_j|}{|U|} \end{aligned}$$

Where P_j is the set of users who gave positive ratings to item j , N_j is the set of users who gave negative ratings to item j , and E_j is the set of users who did not rate item j . Liang et al. defined Improved Jaccard (IJ) in equation (11).

$$\begin{aligned} \text{IJ}(u_1, u_2) &= \left(\sum_{j \in PA} S_j^P + \sum_{j \in NA} S_j^N + \sum_{j \in D} \sqrt{S_j^P S_j^N} \right) \\ &\quad / \left(\sum_{j \in PA} S_j^P + \sum_{j \in NA} S_j^N + \sum_{j \in D} \sqrt{S_j^P S_j^N} + \sum_{j \in PO} \sqrt{S_j^P S_j^E} + \sum_{j \in NO} \sqrt{S_j^N S_j^E} \right) \end{aligned} \quad (11)$$

As a result, when Liang et al. (2015) concerned singularity quantities, they concerned specific features of users so that the recommendation process is more appropriate. For example, in Table 2, given two rating vectors, $u_3 = (s_{31}=4, s_{32}=1, s_{33}=5, s_{34}=5)$ and $u_4 = (s_{41}=1, s_{42}=2, s_{43}=?, s_{44}=?)$ from table 1.

Table 2. An Example for Ratings

	I1	I2	I3	I4
U1	$s_{11} = 1$	$s_{12} = 2$	$s_{13} = 1$	$s_{14} = 5$
U2	$s_{21} = 2$	$s_{22} = 1$	$s_{23} = 2$	$s_{24} = 4$
U3	$s_{31} = 4$	$s_{32} = 1$	$s_{33} = 5$	$s_{34} = 5$
U4	$s_{41} = 1$	$s_{42} = 2$	$s_{43} = ?$	$s_{44} = ?$

We have $PA = \{\}$, $NA = \{1, 2\}$, $D = \{\}$, $PO = \{3, 4\}$, $NO = \{\}$, $P_1=1$, $N_1=3$, $E_1=0$, $P_2=0$, $N_2=4$, $E_2=0$, $P_3=1$, $N_3=2$, $E_3=1$, $P_4=3$, $N_4=0$, and $E_4=1$ when $r_m=3$. We also obtain $S_1^P = 1 - 1/4 = 3/4$, $S_1^N = 1 - 3/4 = 1/4$, $S_1^E = 1 - 0/4 = 1$, $S_2^P = 1 - 0/4 = 1$, $S_2^N = 1 - 4/4 = 0$, $S_2^E = 1 - 0/4 = 1$, $S_3^P = 1 - 1/4 = 3/4$, $S_3^N = 1 - 2/4 = 1/2$, $S_3^E = 1 - 1/4 = 3/4$, $S_4^P = 1 - 3/4 = 1/4$, $S_4^N = 1 - 0/4 = 1$, and $S_4^E = 1 - 1/4 = 3/4$. As a result, the IJ measure of u_3 and u_4 is:

$$IJ(u_3, u_4) = \frac{S_1^N + S_2^N}{S_1^N + S_2^N + \sqrt{S_3^P S_3^E} + \sqrt{S_4^P S_4^E}} = \frac{1/4 + 0}{1/4 + 0 + \sqrt{3/4 * 3/4} + \sqrt{1/4 * 3/4}} \cong 0.1745$$

On the same page, Lee (2017) [31] divided the rating range into three sub-intervals; for each sub-interval, a sub-Jaccard measure is computed. Finally, the whole Jaccard measure of Lee called JacLMH is the average of such three sub-Jaccard measures. Let r_L and r_H be the minimum values and maximum values of rating values; for example, if rating values range from 1 to 5, then $r_L = 1$ and $r_H = 5$. Let L_{bd} and H_{bd} be boundaries of the three sub-intervals such that $r_L < L_{bd} < H_{bd} < r_H$. The set of items rated by user i denoted I_i is divided into three sub-sets, I_{Li} , I_{Mi} , and I_{Hi} as follows (Lee, 2017):

$$\begin{aligned} I_{Li} &= \{j \in I_i: r_{ij} \leq L_{bd}\} \\ I_{Mi} &= \{j \in I_i: L_{bd} < r_{ij} < H_{bd}\} \\ I_{Hi} &= \{j \in I_i: r_{ij} \geq H_{bd}\} \end{aligned}$$

Traditional Jaccard does not concern numerical rating values. As a result, Ayub, Ghazanfar, Khan, & Saleem (2020) [32] improved Jaccard measure by numerical equality. Let $N_T(u_1, u_2)$ be the count of ratings that are equal in numerical values, as given in equation (12).

$$N_T(u_1, u_2) = \begin{cases} |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| + 1 & \text{if } \forall j \in I_1 \cap I_2, r_{1j} = r_{2j} \\ |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| & \text{otherwise} \end{cases} \quad (12)$$

Their proposed measure called RatingJaccard was defined in equation (13):

$$\text{RatingJaccard}(u_1, u_2) = \frac{N_T(u_1, u_2)}{|I_1 \cup I_2|} \quad (13)$$

Ayub et al. (2020) [32] also improved the quantity $N_T(u_1, u_2)$ by associating it with a predefined threshold TH as given in equation (14).

$$N_T(u_1, u_2) = \begin{cases} |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| + 1 & \text{if } \forall j \in I_1 \cap I_2, r_{1j} = r_{2j} \text{ or } |\bar{u}_1 - \bar{u}_2| \leq TH \\ |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| & \text{otherwise} \end{cases} \quad (14)$$

Where the mean values of u_1 and u_2 are denoted by $\bar{u}_1$ and $\bar{u}_2$, respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

Bag, Kumar, & Tiwari (2019) [15] improved Jaccard by considering more un-co-rated items, which are items rated only by one user among two concerned users. According to their viewpoint, the traditional Jaccard is reformulated as given in equation (15):

$$\text{Jaccard}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|I_1| + |I_2| - |I_1 \cap I_2|} \quad (15)$$

They defined $|\bar{I}_1|$ and $|\bar{I}_2|$ is the number of items that are rated by only user 1 and user 2, respectively.

$$|\bar{I}_1| = |I_1| - |I_1 \cap I_2|, |\bar{I}_2| = |I_2| - |I_1 \cap I_2|$$

The traditional Jaccard is rewritten as given in equation (16).

$$\text{Jaccard}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|\bar{I}_1| + |\bar{I}_2| + |I_1 \cap I_2|} = \frac{1}{1 + \frac{|\bar{I}_1| + |\bar{I}_2|}{|I_1 \cap I_2|}} \quad (16)$$

Therefore, the Jaccard measure is proportional to two quantities $(|\bar{I}_1| + |\bar{I}_2|)^{-1}$ and $|I_1 \cap I_2|$. Bag et al. (2019) improved such proportion by specifying that their Jaccard will be proportional to three quantities, $|\bar{I}_1|^{-1}$, $|\bar{I}_2|^{-1}$, and $|I_1 \cap I_2|$. As a result, the relevant Jaccard was defined in equation (17).

$$\text{RelevantJaccard}(u_1, u_2) = \frac{1}{1 + \frac{1}{|I_1 \cap I_2|} + \frac{|\bar{I}_1|}{1 + |\bar{I}_1|} + \frac{1}{1 + |\bar{I}_2|}} \quad (17)$$

Cosine, which is the most popular similarity measure, is specified as given in equation (18).

$$\text{sim}(u_1, u_2) = \cos(u_1, u_2) = \frac{u_1 \cdot u_2}{|u_1||u_2|} = \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2}} \quad (18)$$

where $u_1 \cdot u_2$ is the dot product (scalar product) of u_1 and u_2 , and $|u_1|$ and $|u_2|$ are the lengths of u_1 and u_2 , respectively. The cosine measure has a range of 0 to 1 if every rating is non-negative. Two users are completely different if it is equal to 0. Two users are the same if it is equal to 1. This is why cosine measure is more logical, but it has a flaw: two vectors may have endpoints that are far apart based on Euclidean distance, but their cosine angle is high. This is a drawback of Euclidean distance, which makes cosine similarity less accurate. As a result, an enhanced version called the triangle area (TA) measure [4] was proposed. Figure 1 illustrates the TA measure.

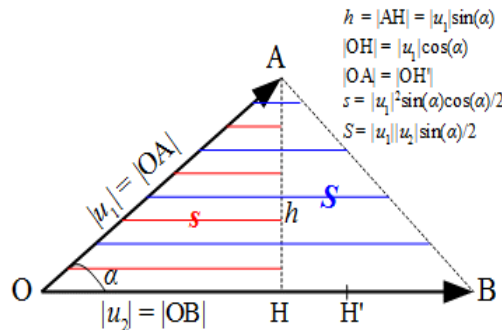


Figure 1. TA measure with $0 \leq \alpha \leq \pi/2$

While maintaining the low complexity and efficacy of both the cosine measure and Euclidean distance in determining the similarity of two vectors, the TA metric uses the ratio of triangle area to the entire triangle

area as a supplementary factor for Euclidean distance to mitigate the adverse impact of Euclidean distance. Equation 19 defines TA, which is regarded as an enhanced cosine measure.

$$\begin{aligned} u_1 \cdot u_2 \geq 0: \text{TA}(u_1, u_2) &= \begin{cases} \frac{(u_1 \cdot u_2)^2}{|u_1|(|u_2|)^3} & \text{if } |u_1| \leq |u_2| \\ \frac{(u_1 \cdot u_2)^2}{(|u_1|)^3|u_2|} & \text{if } |u_1| > |u_2| \end{cases} \\ u_1 \cdot u_2 < 0: \text{TA}(u_1, u_2) &= \begin{cases} \frac{u_1 \cdot u_2}{(|u_2|)^2} & \text{if } |u_1| \leq |u_2| \\ \frac{u_1 \cdot u_2}{(|u_1|)^2} & \text{if } |u_1| > |u_2| \end{cases} \end{aligned} \quad (19)$$

Where, as usual:

$$\begin{aligned} u_1 \cdot u_2 &= \sum_{j \in I_1 \cap I_2} r_{1j} r_{2j} \\ |u_1| &= \sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \\ |u_2| &= \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2} \end{aligned}$$

In addition to cosine, which is given in equation 20, another common similarity metric is Pearson correlation.

$$\begin{aligned} \text{Pearson}(u_1, u_2) &= \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{u}_2)^2}} \end{aligned} \quad (20)$$

Where the mean values of u_1 and u_2 are denoted by $\bar{u}_1$ and $\bar{u}_2$, respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

The Pearson measure has a spectrum of -1 to 1 . Two users are completely at odds if it is equivalent to -1 . Two users are the same if it is equivalent to 1 . In statistics, the sample correlation coefficient is known as the Pearson measure. Choi and Suh [33] proposed the weighted Pearson measure, which was weighted by the similarities ratio of items. Each factor in the PC measure is weighted by a similarity of an active item and another item. Suppose it is necessary to estimate rating values of active item k ; the PC measure is defined in equation 21.

$$\text{PC}_k(u_1, u_2) = \frac{\sum_{j \in I} \left(\left(\text{sim}(v_k, v_j) \right)^2 (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2) \right)}{\sqrt{\sum_{j \in I_1} \left(\text{sim}(v_k, v_j) (r_{1j} - \bar{u}_1) \right)^2} \sqrt{\sum_{j \in I_2} \left(\text{sim}(v_k, v_j) (r_{2j} - \bar{u}_2) \right)^2}} \quad (21)$$

Where $\text{sim}(v_k, v_j)$ is the similarity of the taken items k and j . Note, $\text{sim}(v_k, v_j)$ can be calculated by any measures here.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}$$

$$\bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

On the other hand, to enhance recommendation tasks with limited ratings, Liu, Hu, Mian, Tian, & Zhu (2014) [8] introduced a similarity measure known as NHMS. The sigmoid function and the enhanced PIP measure, known as PSS (Proximity – Significance, Singularity), served as the foundation for the NHMS measure. Equation (22) calculates PSS similarity.

$$PSS(u_1, u_2) = \sum_{j \in I} \text{Proximity}(r_{1j}, r_{2j}) * \text{Significance}(r_{1j}, r_{2j}) * \text{Singularity}(r_{1j}, r_{2j}) \quad (22)$$

Where, $I = I_1 \cap I_2$ is the intersection set of I_1 and I_2 . The proximity factor determines the similarity of two ratings, based on the distance between them. The similarity between two evaluations is determined by the significance factor, based on the distance from them to the rating median, with small distance reflecting the higher similarity.

$$\text{Proximity}(r_{1j}, r_{2j}) = 1 - \frac{1}{1 + \exp(-|r_{1j} - r_{2j}|)}$$

$$\text{Significance}(r_{1j}, r_{2j}) = \frac{1}{1 + \exp(-|r_{1j} - r_m| |r_{2j} - r_m|)}$$

$$\text{Singularity}(r_{1j}, r_{2j}) = 1 - \frac{1}{1 + \exp\left(-\left|\frac{r_{1j} + r_{2j}}{2} - \mu_j\right|\right)}$$

Based on how two ratings differ from one another, the significance factor establishes how comparable they are. The sigmoid function-based equations for these factors are as follows. Note that whereas μ_j is the rating mean of item j , r_m is the median of the rating values. For instance, if the rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$. The similarity between two users was also taken into account by Liu et al. (2014) using the URP measure, which is provided in equation 23.

$$\text{URP}(u_1, u_2) = 1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)} \quad (23)$$

Where μ_1 and μ_2 represent the rating average of users 1 and 2, respectively, and σ_1 and σ_2 represent their rating standard deviations.

$$\mu_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \mu_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

$$\sigma_1 = \sqrt{\frac{1}{|I_1|} \sum_{j \in I_1} (r_{1j} - \mu_1)^2}, \sigma_2 = \sqrt{\frac{1}{|I_2|} \sum_{j \in I_2} (r_{2j} - \mu_2)^2}$$

Lastly, as shown in equation 24, Liu et al. (2014) presented a new heuristic similarity model (NHSM) as the triple product of PSS, URP, and Jaccard2 measures.

$$\text{NHSM}(u_1, u_2) = \text{PSS}(u_1, u_2) * \text{URP}(u_1, u_2) * \text{Jaccard2}(u_1, u_2) \quad (24)$$

Traditional metrics like cosine and Pearson have the disadvantage of being reliant on shared items that two interested individuals scored. Therefore, Deng et al. (2018) [34] proposed the similarity model, which does not require such commonly rated items, as given in equation 25.

$$\text{sim}(u_1, u_2) = \sum_{j \in I_1} \sum_{k \in I_2} \text{sim}_{\text{item}}(j, k) * s(r_{1j}, r_{2k}) \quad (25)$$

The quantity $s(r_{1j}, r_{2k})$ is called the local similarity function based on the sigmoid function, which is defined in equation 26.

$$s(r_{1j}, r_{2k}) = \frac{1}{1 + \exp\left(- (r_{1j} - \bar{u}_1)(r_{2k} - \bar{u}_2)\right)} \quad (26)$$

Where the mean values of u_1 and u_2 are denoted by $\bar{u}_1$ and $\bar{u}_2$, respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

The quantity $\text{sim}_{\text{item}}(j, k)$ is the similarity measure of item j and item k , which does not require common (co-rating) users. Deng et al. [34] applied Kullback-Leibler (KL) divergence into calculating the similarity $\text{sim}_{\text{item}}(j, k)$ as given in equation 27.

$$\text{sim}_{\text{item}}(j, k) = \sum_{v=r_{\min}}^{r_{\max}} p_{jv} \log \frac{p_{jv}}{p_{kv}} \quad (27)$$

Note, $r_{\min}$ and $r_{\max}$ are the lowest and highest values of ratings, respectively. Where p_{jv} (p_{kv}) is the probability of rating value v that item j (item k) received among its ratings, as given in equations 28-29.

$$p_{jv} = \frac{|u \in U: r_{uj} = v|}{|u \in U: r_{uj} \text{ not missing}|} \quad (28)$$

$$p_{kv} = \frac{|u \in U: r_{uk} = v|}{|u \in U: r_{uk} \text{ not missing}|} \quad (29)$$

To further enhance CF performance, SMD was introduced by Amer et al. (2021) in [16]. Given two rating vectors of U_1 and U_2 , respectively, with some r_{ij} absent (empty), $u_1 = (r_{11}, r_{12}, \dots, r_{1n})$ and $u_2 = (r_{21}, r_{22}, \dots, r_{2n})$. If r_{ij} is not present (rated), it is changed to 1 in binary format; if not, it is converted to 0. Let N_{12} be the total number of "1" values that are shared by u_1 and u_2 . In this instance, $N = n$. Let N be the aggregate number of all items. Let N_1 and N_2 be the values of "1" for u_1 and u_2 , correspondingly. Let F be the entire number of differences between u_1 and u_2 . For instance, $r_{11} = 0$ and $r_{21} = 1$ add one difference to F . Amer et al. (2021) defined the SMD as given in equation 30.

$$\text{SMD} = \frac{\left(1 - \frac{F}{N}\right) + \left(\frac{2N_{12}}{N_1 + N_2}\right)}{2} \quad (30)$$

Let I_1 or I_2 be sets of items rated by users 1 and user 2, respectively. Generally speaking, measure SMD was typically constructed as weight vectors of users/items, where each vector element is a weight that is the product of term frequency (TF) and inverse document frequency (IDF), yet they are used in CF as shown in the next example. Given two rating vectors, $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$. These two have binary representations that are (1, 1, 1, 1, 0, 1) and (1, 0, 0, 1, 1, 1). According to the mechanism of SMD, we have $N_{12}=3$, $N=6$, $F=3$, $N_1=5$, and $N_2=4$. Therefore, the measure is computed as given below.

$$\text{SMD} = \frac{\left(1 - \frac{3}{6}\right) + \left(\frac{2 * 3}{5 + 4}\right)}{2} = 0.58$$

Absolute Difference of Ratings (ADR) was defined by Gazdar & Hidri (2019) [32], as given in equation 31.

$$\text{ADR}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \exp\left(-\frac{|r_{1j} - r_{2j}|}{\max(r_{1j}, r_{2j})}\right)}{|I_1 \cap I_2|} \quad (31)$$

Note, $\max(r_{1j}, r_{2j})$ is the maximum value among r_{1j} and r_{2j} . Jin, Zhang, Cai, & Zhang (2020) [36] combined the local similarity and global similarity with the support of the singularity factor. As a result, they proposed a combined Local and Global (CLAG) measure as drawn in equation 34. Let r_m be the median of the rating values. Given user 1 and user 2, for every item j , let:

- If both r_{1j} and r_{2j} are positive, item j belongs to the predefined set PA .
- If both r_{1j} and r_{2j} are negative, item j belongs to the predefined set NA .
- If r_{1j} and r_{2j} are different, which means that one is positive and the other is negative, item j belongs to the predefined set D .

Let S_j^P and S_j^N be positive singularity and negative singularity.

$$S_j^P = 1 - \frac{|P_j|}{|U|}$$

$$S_j^N = 1 - \frac{|N_j|}{|U|}$$

Where P_j is the collection of individuals who provided positive ratings to item j and N_j is the collection of individuals who provided negative ratings to item j . Zhang et al. (2020) defined their singularity measure, as given in equations 32-33.

$$\text{singularity}(u_1, u_2) = \frac{1}{3} (L_P(u_1, u_2) + L_N(u_1, u_2) + L_D(u_1, u_2)) \quad (32)$$

Where,

$$L_P(u_1, u_2) = \frac{1}{|PA|} \sum_{j \in PA} \left(1 - \frac{1}{1 + \exp(-|r_{1j} - r_{2j}|)}\right) (S_j^P)^2$$

$$L_N(u_1, u_2) = \frac{1}{|NA|} \sum_{j \in NA} \left(1 - \frac{1}{1 + \exp(-|r_{1j} - r_{2j}|)}\right) (S_j^N)^2$$

$$L_D(u_1, u_2) = \frac{1}{|D|} \sum_{j \in D} \left(1 - \frac{1}{1 + \exp(-|r_{1j} - \bar{u}_1||r_{2j} - \bar{u}_2|)}\right) S_j^P S_j^N \quad (33)$$

Note, $\bar{u}_i$ is mean of u_i .

$$\bar{u}_i = \frac{1}{|I_i|} \sum_{j \in I_i} r_{ij}$$

Mean squared difference (MSD) is defined as the reciprocal of the distance between two vectors as given in equation 34. Let MAX be the highest rating value. MSD is computed as follows [8].

$$\text{MSD}(u_1, u_2) = 1 - \frac{\sum_{j \in I} \left(\frac{r_{1j} - r_{2j}}{\text{MAX}}\right)^2}{|I|} \quad (34)$$

Feng et al. [14] presented a new similarity model with three components to address the sparse rating matrix problem, S_1 , S_2 , and S_3 . The S_1 is the cosine similarity in equation 35.

$$S_1(u_1, u_2) = \begin{cases} \text{cosine}(u_1, u_2) & \text{if sparsity} < \rho \\ \text{COJ}(u_1, u_2) & \text{otherwise} \end{cases} \quad (35)$$

Where ρ is the sparsity threshold that Feng et al. suggested. User pairs with few co-rated products are penalized by the S_2 , as given in equation 36.

$$S_2(u_1, u_2) = \frac{1}{1 + \exp\left(-\frac{|I_1 \cap I_2|^2}{|I_1||I_2|}\right)} \quad (36)$$

The statistical aspect of user ratings, which represent key user preferences, is the main focus of the S_3 . The previously mentioned URP metric is S_3 , as given in equation 37.

$$S_3(u_1, u_2) = \text{URP}(u_1, u_2) = 1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)} \quad (37)$$

In this case, μ_1 and μ_2 represent the rating average of users 1 and 2, correspondingly, and σ_1 and σ_2 represent their rating standard deviations. Feng's similarity model is the following product of S_1 , S_2 , and S_3 , as drawn in equation 38.

$$\text{Feng}(u_1, u_2) = S_1(u_1, u_2) * S_2(u_1, u_2) * S_3(u_1, u_2) \quad (38)$$

The rating matrix is viewed by Chen et al. (Chen, Zhang, Liu, Gao, & Zhou, 2016) [38] as a user-item bipartite network, where each connection in the network reflects the rating that a user assigned to an item. "The resource-allocation (RA) method is equal to the one-step random walk approach in the user-object bipartite network originating from the shared neighbors," according to Chen et al. As a result, the RA index between users 1 and 2 is defined in equation 39.

$$\text{RA}(u_1, u_2) = \sum_{j \in I_1 \cap I_2} \frac{r_{1j} r_{2j}}{|v_j|^2} \quad (39)$$

Where $|v_j|$ is module of item j ,

$$|v_j| = \sqrt{\sum_{i \in J_j} r_{ij}^2}$$

Remember that J_j is the group of users who gave item j a rating. Although Chen et al. actually computed the RA index between two items in the original research, their concept may be used to both item-based and user-based NN algorithms.

Lee (2018) [39] applied information entropy to metrics like cosine and Pearson to enhance similarity measures. The entropy of item j , called $r(j)$, is expressed in equation 40. Let $r_{\min}$ and $r_{\max}$ be the lowest and highest values of ratings, and let $r(j)$ be the rating for item j (individuals who rated such item j are not yet involved).

$$E(j) = - \sum_{v=r_{\min}}^{r_{\max}} P(r(j) = v) \log_2(P(r(j) = v)) \quad (40)$$

Where $P(r(j)=v)$ is the probability that users rated item j with rating value v , as given in equation 41.

$$P(r(j) = v) = \frac{|u \in \mathbf{U}: r_{uj} = v|}{|u \in \mathbf{U}: r_{uj} \text{ not missing}|} \quad 41$$

Where r_{uj} is the rating value that user u did rate on item j . Lastly, Lee [30] devised the entropy Pearson by connecting the entropy to Pearson as given in equation 42.

$$\text{pearson}_e(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2)E(j)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{u}_2)^2}} \quad (42)$$

In the meantime, three additional measures, cosine, Jaccard, and MMD, were merged by CjacMD [9], as given in equation 43.

$$\text{CjacMD} = \cos(u_1, u_2) + \text{Jaccard}(u_1, u_2) + \text{MMD}(u_1, u_2) \quad (43)$$

According to Harris & Sjøvold (2018) [40], the MMD measure is defined in equation 44.

$$\text{MMD}(u_1, u_2) = \frac{1}{1 + \frac{1}{b} \sum_{j=1}^b \left((\theta_{1j} - \theta_{2j})^2 - \frac{1}{0.5 + x_j} - \frac{1}{0.5 + y_j} \right)} \quad (44)$$

Where Grewal's transformations of X and Y are denoted by θ_{1*} and θ_{2*} , respectively.

$$\theta_{1j} = \frac{1}{\sin \left(1 - \frac{2x_j}{|I_1|} \right)}$$

$$\theta_{2j} = \frac{1}{\sin \left(1 - \frac{2y_j}{|I_2|} \right)}$$

Additionally, Sun et al. (2017) [10] devised a new measure known as the Triangle multiplying Jaccard (TMJ) measure by combining the Triangle and Jaccard measures. The definition of the integrated TMJ is defined in equation 45.

$$\text{TMJ}(u_1, u_2) = \text{Triangle}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (45)$$

Where,

$$\text{Triangle}(u_1, u_2) = 1 - \frac{|AB|}{|OA| + |OB|} = 1 - \frac{|u_1 - u_2|}{|u_1| + |u_2|} = 1 - \frac{\sqrt{\sum_{j \in I} (r_{1j} - r_{2j})^2}}{\sqrt{\sum_{j \in I} r_{1j}^2} + \sqrt{\sum_{j \in I} r_{2j}^2}}$$

Gazdar and Hidri (2019) [26] suggested the OS measure when considering the kernel function to better improve CF accuracy. The following is how their OS is calculated using the Absolute Difference of Ratings (ADR) and the Percentage of Non-Common Ratings (PNCR), as given in equation 46.

:

$$\text{OS}(u_1, u_2) = \text{PNCR}(u_1, u_2) * \text{ADR}(u_1, u_2) \quad (46)$$

Where (Gazdar & Hidri, 2019, p. 19),

$$\text{PNCR}(u_1, u_2) = \exp\left(-\frac{|V| - |I_1 \cap I_2|}{|V|}\right)$$

$$\text{ADR}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \exp\left(-\frac{|r_{1j} - r_{2j}|}{\max(r_{1j}, r_{2j})}\right)}{|I_1 \cap I_2|}$$

Keep in mind that $\max(r_{1j}, r_{2j})$ is the highest value between r_{1j} and r_{2j} , and $|V|$ is the total number of items. Furthermore, numbers $\text{PNCR}(u_1, u_2)$ and $\text{ADR}(u_1, u_2)$ can be regarded as separate measurements of similarity. Using the statistical idea of "statistical ratio match," Verma & Aggarwal (2019) [41] introduced the Simple Matching Coefficient (SMC) measure. Two rating values, r_{1j} and r_{2j} , match together in the following three scenarios:

1. They are equal as $r_{1j} = r_{2j}$.
2. Both of them are smaller than r_m as $r_{1j} < r_m$ and $r_{2j} < r_m$.
3. Both of them are larger than r_m as $r_{1j} > r_m$ and $r_{2j} > r_m$.

Consequently, SMC was defined by Verma and Aggarwal as the following ratio of matched items to total items (see equation 47).

$$\text{SMC}(u_1, u_2) = \frac{|\{j \in I_1 \cap I_2 : r_{1j} \text{ matches } r_{2j}\}|}{|I_1 \cap I_2|} \quad 47$$

However, Tan and He (2017) [42] developed a resonance similarity (RES) measure by drawing inspiration from the physical resonance of simple harmonic motion. The basic harmonic motion of particles in a vibration system is thought to be comparable to user rating behavior. Three components, consistency, distance, and Jaccard, were combined to form RES, as drawn in equation 48.

$$\text{RES}(u_1, u_2) = \left(\sum_{j \in I_1 \cap I_2} C(u_1, u_2, v_j, k_1) D(u_1, u_2, v_j, k_2, k_3) \right) J(u_1, u_2, k_4) \quad (48)$$

Note, v_j represents item j while k_1, k_2, k_3 , and k_4 are predefined constants. Finally, the modified PIP (MPIP) measure was proposed by Manochandar and Punniyamoorthy (2020) [17]. This measure remains the total of the products of the triples Proximity, Impact, and Popularity, but it has been enhanced. For example, the following is how the Proximity quantity is defined in equation 49.

$$\text{Proximity2}(r_{1j}, r_{2j}) = \begin{cases} \left(\frac{|r_{1j} - r_{2j}| - \frac{r_m^+ + r_m^-}{2}}{r_{max} - r_{min}} \right)^2 & \text{if agree}(r_{1j}, r_{2j}) = \text{true} \\ \delta \left(\frac{1}{\frac{|r_{1j} - r_{2j}|}{r_{max} - r_{min}}} \right)^2 & \text{if agree}(r_{1j}, r_{2j}) = \text{false} \end{cases} \quad 49$$

By similar inference, r_m^+ is the specified average that is larger than r_m ; for example, if rating values range from 1 to 5, then, $r_m^+ = (4+5) / 2 = 4.5$. Thus, r_m^- is the specified average that is smaller than r_m ; for example, if rating values range from 1 to 5, then, $r_m^- = (1+2) / 2 = 1.5$. The parameter δ is a rating scale as follows:

$$\delta = \begin{cases} 0.75 & \text{if } |r_{1j} - r_{2j}| > r_m \\ 0.5 & \text{if } |r_{1j} - r_{2j}| = r_m \\ 0.25 & \text{otherwise} \end{cases}$$

The Impact quantity was changed by Manochandar and Punniyamoorthy as given in equation 50.

$$\text{Impact2}(r_{1j}, r_{2j}) = \begin{cases} \exp \left(- \frac{1}{(|r_{1j} - r_m| + 1) * (|r_{2j} - r_m| + 1)} \right) & \text{if agree}(r_{1j}, r_{2j}) = \text{true} \\ \frac{1}{(|r_{1j} - r_m| + 1) * (|r_{2j} - r_m| + 1)} & \text{if agree}(r_{1j}, r_{2j}) = \text{false} \end{cases} \quad 50$$

Additionally, they made the following changes to the Popularity quantity, as drawn in equation 51.

$$\text{Popularity2}(r_{1j}, r_{2j}) = \begin{cases} \log_{10} \left(2 + \left(\frac{r_{1j} + r_{2j}}{2} - \mu_j \right)^2 \right) & \text{if } (r_{1j} > \mu_j \text{ and } r_{2j} > \mu_j) \\ & \text{or } (r_{1j} < \mu_j \text{ and } r_{2j} < \mu_j) \\ 0.3010 & \text{otherwise} \end{cases} \quad 51$$

Note, μ_j is the average rating of item j . As a result, MPIP is finally defined in equation 52.

$$\text{MPIP}(u_1, u_2) = \sum_{j \in I_1 \cap I_2} \text{Proximity2}(r_{1j}, r_{2j}) * \text{Impact2}(r_{1j}, r_{2j}) * \text{Popularity2}(r_{1j}, r_{2j}) \quad 52$$